



SIMATS
ENGINEERING



SIMATS
Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

AI-Powered Resume Screening and Candidate Ranking System

CSA0974-JAVA PROGRAMMING FOR DEVELOPERS

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INTRODUCTION



- This project is a web-based application that automates the process of screening resumes and selecting suitable candidates for job openings.
- The system uses Artificial Intelligence (AI) and Natural Language Processing (NLP) to extract skills, qualifications, and experience from uploaded resumes.
- It compares candidate profiles with job requirements and generates a score to rank applicants based on their suitability.
- The system reduces manual effort, saves recruitment time, improves hiring accuracy, and helps recruiters make better decisions.

METHODOLOGY

- **Resume Collection and Upload**

Candidates upload their resumes in PDF or DOCX format through the web application, and the system stores them securely in the database.

- **Information Extraction**

AI and NLP techniques are used to extract important details such as skills, education, experience, certifications, and keywords from the resumes.

- **Job Matching and Scoring**

The extracted information is compared with job requirements, and a matching score is calculated based on skill and qualification relevance.

- **Candidate Ranking and Reporting**

Candidates are ranked according to their scores, and recruiters can view the results through a dashboard and generate reports for hiring decisions.



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LITERATURE REVIEW:

- Traditional resume screening is a manual process that takes a lot of time and effort.
- AI helps automate the screening process by analyzing resumes quickly.
- Natural Language Processing (NLP) is used to extract skills, education, and experience from resumes.
- Machine Learning algorithms compare candidate profiles with job requirements.
- The system ranks candidates based on their suitability for a job position.

SYSTEM IMPLEMENTATION

- **User Interface Layer**

Recruiters and candidates interact with the system through a web interface for login, resume upload, job posting, and viewing results.

- **Application Layer (Spring Boot)**

Handles business logic, user authentication, resume processing, candidate ranking, and communication between the frontend and database.

- **AI/NLP Processing Layer**

Extracts skills, education, experience, and keywords from resumes and matches them with job requirements to generate candidate scores.

- **Database Layer (MySQL/PostgreSQL)**

Stores user information, resumes, job postings, extracted data, candidate scores, and recruitment reports securely.

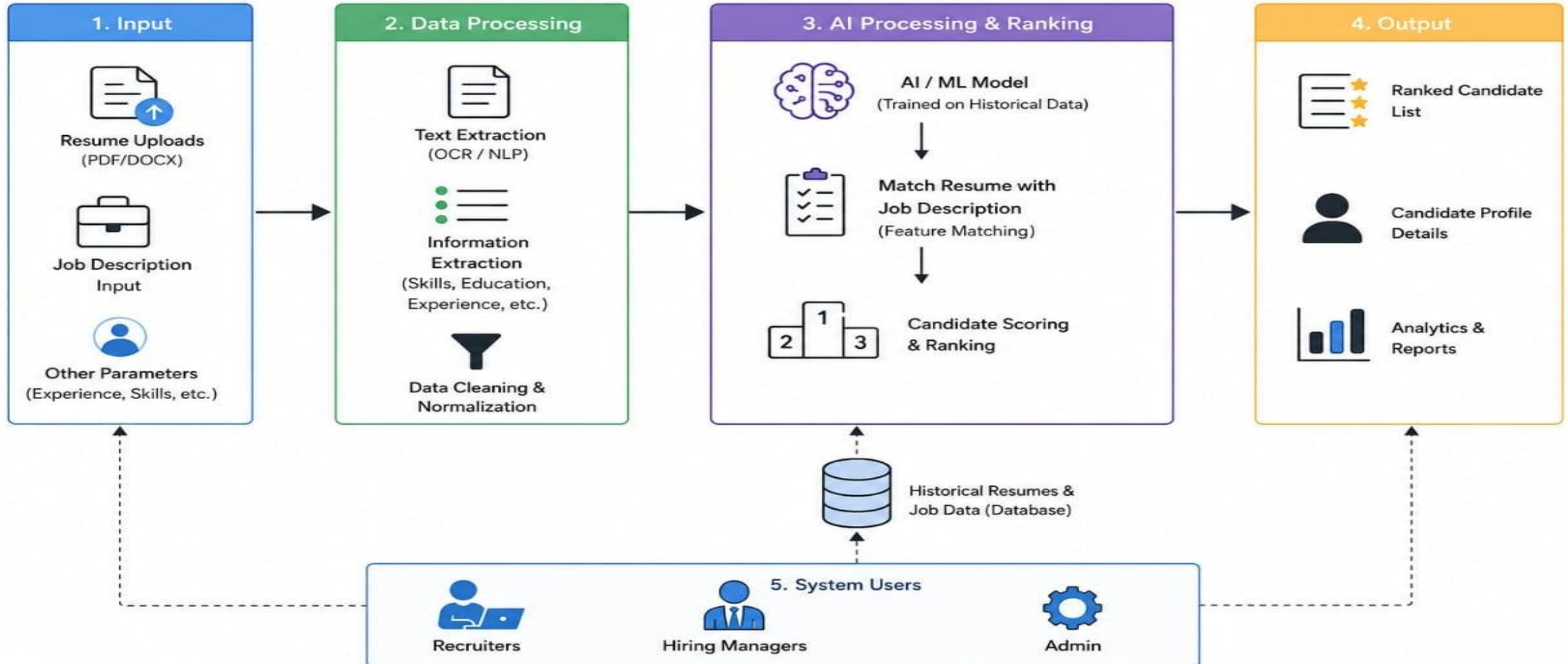


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SYSTEM ARCHITECTURE:

AI-Powered Resume Screening and Candidate Ranking System – Architecture





MODULE 1- Resume Processing and Skill Extraction



- **Resume Upload**

Candidates can upload resumes in PDF or DOCX format

- **Resume Parsing**

The system extracts text and relevant information from uploaded resumes automatically.

- **Skill Extraction**

AI/NLP techniques identify technical skills, soft skills, certifications, and qualifications.

- **Experience & Education Analysis**

The system extracts candidate experience, educational background, and achievements.

- **Data Storage**

Extracted information is stored in the database for further matching and ranking processes.

MODULE 1- Resume Processing and Skill Extraction



**RecruitOS**

AI

DashboardJob PostingsCandidates**AI Screening**Analytics

AI Engine Online

AI Screening Engine

Screening Configuration

TARGET POSITION

Senior Full Stack Engineer
Engineering

ML / AI Engineer
Data Science

DevOps / Platform Engineer
Infrastructure

Product Designer (UX/UI)
Design

SCORING WEIGHTS

Skill Match 70%

Experience 30%

AI Processing Log

100%

```
[00] Initializing AI screening engine...
[01] Loading job description: Senior Full Stack Engineer
[02] Identified 8 required skills, 4 bonus skills
[03] Parsing candidate resume pool...
[04] [1/8] Processing: Arjun Mehta
[05] [2/8] Processing: Priya Sharma
[06] [3/8] Processing: Rohan Verma
[07] [4/8] Processing: Kavya Nair
[08] [5/8] Processing: Siddharth Joshi
[09] [6/8] Processing: Ananya Reddy
[10] [7/8] Processing: Vikram Singh
[11] [8/8] Processing: Meera Iyer
[12] Running skill extraction model...
[13] Computing semantic similarity scores...
[14] Applying experience weighting algorithm...
[15] Generating candidate rankings...
[16] ✓ AI screening complete!
```




MODULE 2-Candidate Screening and Matching



- **User Registration**

Allows recruiters and candidates to create accounts and access the system.

- **User Login & Authentication**

Provides secure login using username/email and password verification.

- **Profile Management**

Users can update personal details, contact information, and account settings.

- **Job Posting Management**

Recruiters can create, edit, view, and delete job postings with required skills and qualifications.

- **Role-Based Access Control**

Different access permissions are provided for recruiters and candidates to ensure system security.

MODULE 2-Candidate Screening and Matching



 **RecruitOS**

AI

Dashboard

Job Postings

Candidates

AI Screening

Analytics

AI Engine Online

REACT

NODE.JS

TYPESCRIPT

POSTGRESQL

SJ

2

Siddharth Joshi

Backend Developer · Hyderabad, India

Skill Match59%

Experience38%

NODE.JS

TYPESCRIPT

POSTGRESQL

DOCKER

REACT

AWS

VS

3

Vikram Singh

Cloud Architect · Gurgaon, India

Skill Match33%

Experience100%

AWS

DOCKER

REACT

NODE.JS

RV

4

Rohan Verma

Skill Match

Experience

Vikram Singh

Cloud Architect

Gurgaon, India

8 yrs exp

53

SUMMARY

Cloud architect with 8 years designing resilient multi-cloud infrastructure. AWS Solutions Architect Professional certified.

MATCH BREAKDOWN

Overall Score53%

Skill Match33%

Experience Match100%

SKILLS VS REQUIREMENTS

✓ Matched Required

AWS

DOCKER



MODULE 3-Candidate Ranking and Reporting



- **Candidate-Job Matching**

The system compares extracted resume data with job requirements to find suitable candidates.

- **Resume Scoring**

Candidates are assigned scores based on skills, qualifications, experience, and keyword matching.

- **Candidate Ranking**

Applicants are ranked in descending order according to their matching scores.

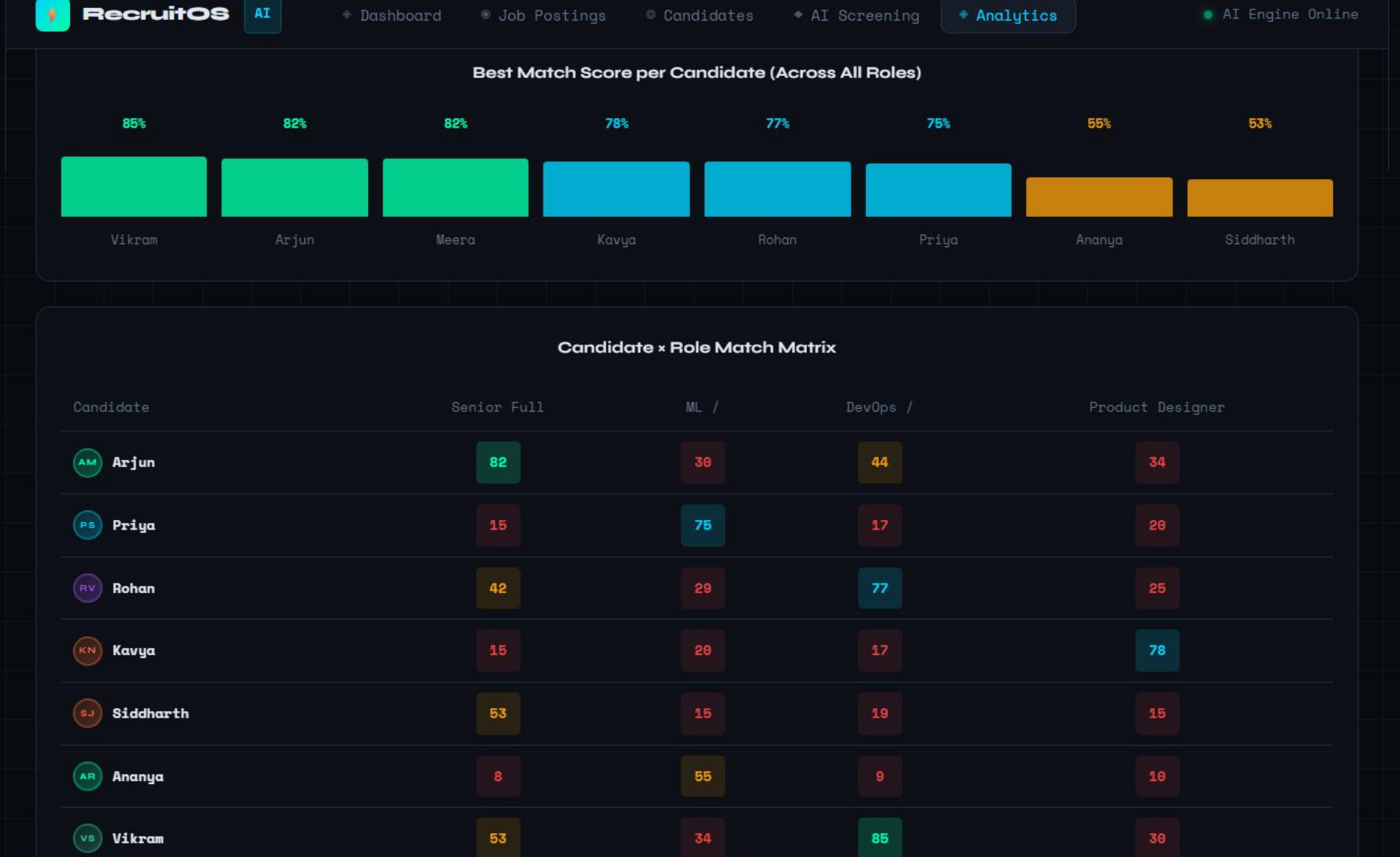
- **Recruiter Dashboard**

Recruiters can view ranked candidate lists and analyze applicant details through an interactive dashboard.

- **Report Generation**

The system generates reports on candidate rankings, hiring statistics, and recruitment performance.

MODULE 3-Candidate Ranking and Reporting



CONCLUSION

- The system automates the resume screening process, reducing manual effort and recruitment time.
- AI and NLP techniques help extract candidate skills, qualifications, and experience accurately.
- Candidate-job matching and ranking improve the quality and efficiency of hiring decisions.
- The system provides a smart, scalable, and reliable solution for modern recruitment and talent acquisition.

FUTURE SCOPE

- Integration with real AI/NLP models for advanced resume analysis.
- Automatic PDF and DOCX resume parsing.
- Real-time job recommendation system for candidates.
- Interview scheduling and notification management.
- Integration with job portals such as LinkedIn and Indeed.
- AI-based interview question generation.



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THANK YOU